

Section: **4.2 – FATIGUE SPECTRUM
GENERATION**

Chapter: **4210**

Title: **THEORETICAL SPECTRA**

Revision: **0.1**

Date: **15/07/2020**

1 INTRODUCTION

This section describes the general methodology used in Airbus D&S for developing theoretical stress spectra, calculated from fatigue loads, to be used in Fatigue and Damage Tolerance analysis.

1.1 METHOD-SUMMARY

Fatigue loads shall represent variety of loading within normal operation of an aircraft. These loads comprise all steady variations and their disturbances expected to occur during the service life of the aircraft.

In general, there are two basic load conditions contributing to the fatigue load definition:

- Steady (one g) loads in various flight stages and standing on ground.
- Incremental loads (disturbances) to the steady condition, due to:
 - o Gusts: Vertical, lateral and/or combined
 - o Flight manoeuvres: Symmetric, coordinated turns and rolling
 - o Military operations: Aerial delivery, refuelling, terrain and collision avoidance, etc.
 - o Ground manoeuvres: Ground roll (taxi, take-off and landing roll), braking, turning, pivoting, towing, etc.
 - o Manoeuvres between flight and ground: Take-off rotation, landing touch down (including spin up and spring back).
 - o Miscellaneous: Wind milling, engine failure, engine run-up, etc.

All above load events are mainly defined by their statistical data, called basic spectra. They allow predictions about distributions of occurrences, needed to transform the fatigue unit loads into load spectra or stress spectra.

Criteria and data included in this manual for theoretical stress spectra development are for main components. For others structural elements like, movable surfaces, landing gear doors, etc., different considerations can be done.

The typical flow diagram to develop discrete stress spectra for an element in each flight segment is shown below:

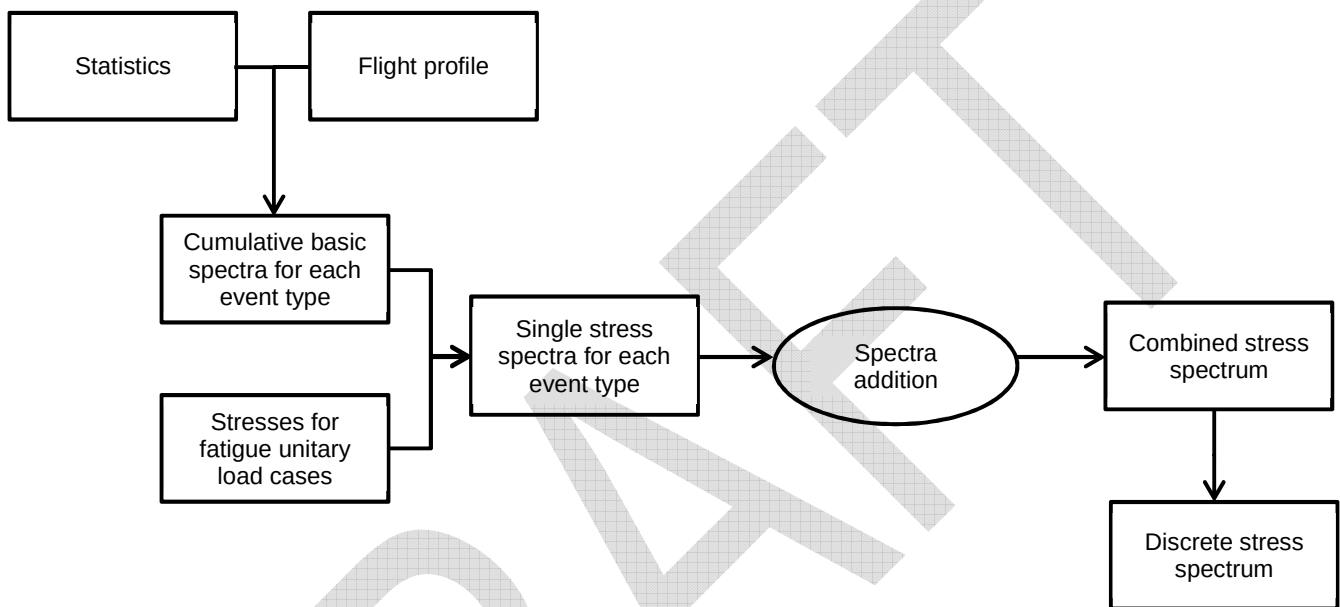


Figure 1. Flow diagram for stress spectra generation

1.2 SCOPE AND LIMITATIONS

The scope of this section is to provide the methodology for stress spectra generation, applicable to transport aircrafts, tankers and, in general, every kind of aircraft for which an utilisation based on standard flight profiles can be stated.

For this reason, this methodology here described does not apply to combat aircrafts, like Eurofighter or Tornado. In these cases, it is recommended to develop global sequences, in terms of load cases, and convert them to stress sequences.

2 METHOD DESCRIPTION

The theoretical stress spectrum is defined as a block spectrum, corresponding to all types of mission and flights in each one of them for the entire DSG of the aircraft.

Therefore, the theoretical spectrum will be split in different missions, each mission containing all its flight segments defined in its flight profile. Each flight segment will contain all stress cycles expected to occur in that segment during the DSG.

The organisation of this spectrum is schematically as follows:

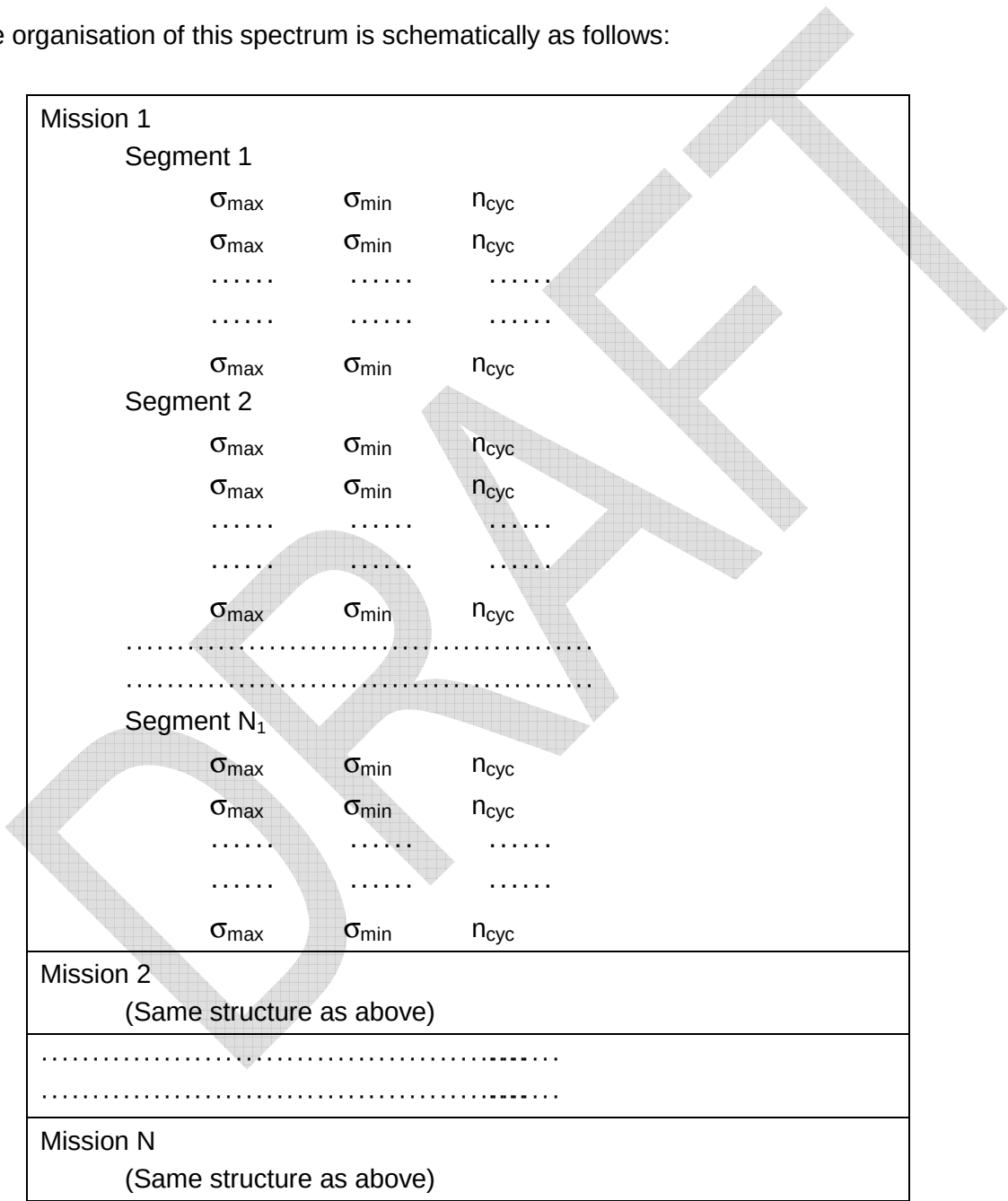


Figure 2. Stress spectrum organisation

2.1 BASIC SPECTRA

In each flight segment of the flight profile of each type of mission, basic spectra will be developed for the considered disturbances from the steady condition, being relevant for the structural component under study. The basic spectra are developed taking as input the statistics and the aircraft flight profiles.

These spectra will be expressed in terms of occurrences during a flight segment (N) of the significant magnitude defining the corresponding load event.

As an example, the following magnitudes are going often to be considered in the basic spectra:

Segment	Load event	Considered magnitude
Flight	Vertical / lateral / combined gust	V_e Equivalent gust speed
	Vertical manoeuvre	Δn_z Increment of the vertical load factor
	Coordinated turn	Δn_z Increment of the vertical load factor or F_{UTF} Factor on UTF (unitary turn for fatigue)
Ground roll	Taxi bumps	Δn_z Increment of the vertical load factor and n_y Lateral load factor
	Turns	n_y Lateral load factor
Landing	Touch down	V_z Sinking speed

Table 1. Example of considered magnitudes for basic spectra

The basic spectra are usually obtained in cumulative form. Each point represents the number of times that the value of the magnitude is reached or exceeded. In general, they will be different for each flight segment into which the flight profile has been divided. Next figure shows an example of a basic cumulative spectrum:

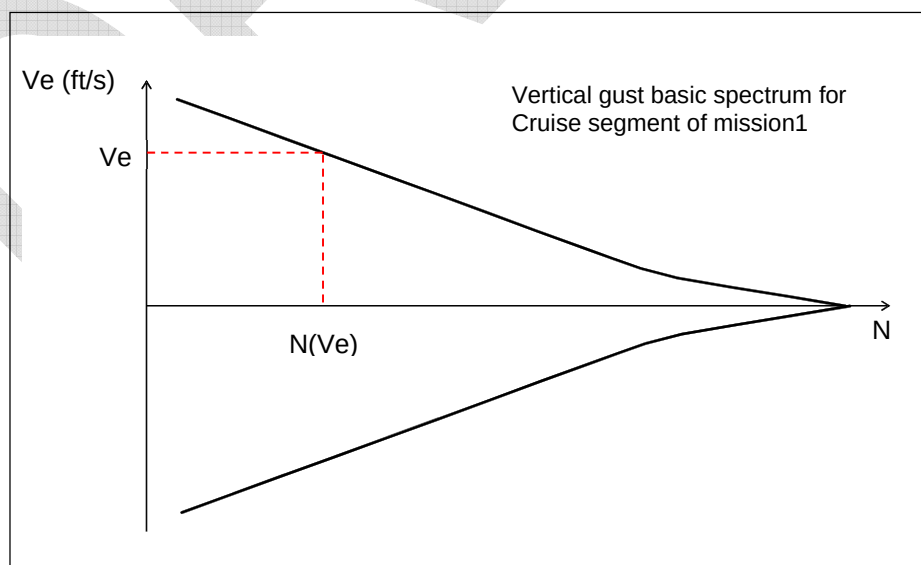


Figure 3. Example of cumulative basic gust spectrum

Some sources where the basic spectra are defined can be found in Ref. 1 and in its included references (here also included as Ref. 2, Ref. 3, Ref. 4 and Ref. 5), being this reference a significant part of the methodology used in Airbus Defence and Space.

2.2 SINGLE STRESS SPECTRUM

Each basic spectrum can be converted to stress spectrum by applying the relationship between the stress in the considered structural element and the basic magnitude, adding finally the stress corresponding to the steady condition.

$$\sigma = \sigma_{1g} + \Delta\sigma$$

$$\Delta\sigma = \frac{\delta\sigma}{\delta B} \cdot B$$

Where σ is the total stress
 $\Delta\sigma$ is the incremental stress, due to the magnitude of the disturbance
 B is the basic magnitude (V_e , Δn_z , F_{UTF} , etc.)

In the example of Figure 3, the cumulative stress spectrum for the selected element for which the steady and the incremental stresses have been calculated, would be:

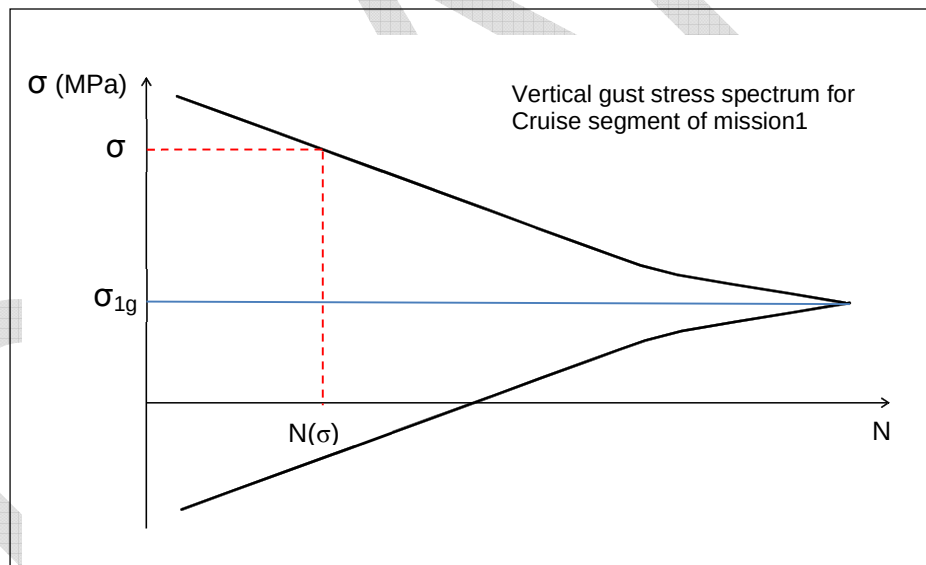


Figure 4. Example of cumulative basic gust spectrum

Taking into account the elastic nature of the aeronautical structures, the incremental stress due to a basic disturbance, can be considered, in general, as a linear function of the basic magnitude. For example, for gust spectra:

$$\Delta\sigma = \frac{\delta\sigma}{\delta V_e} \cdot V_e$$

$$\frac{\delta\sigma}{\delta V_e} = \text{constant}$$

Therefore, the analysis of only one load case (empirically, from FEM or from test), plus the steady condition, is enough to convert a basic spectrum into a stress spectrum, for:

- One selected element
- Only one flight or ground segment
- One basic spectrum

Nevertheless, special considerations about non linearity, combination of loads, etc., have to be accounted for. Some of them are the following:

2.2.1 Non-linear response

Some cases and due to the nature of the involved elements in the loading, the relationship between the stress and the basic magnitude can be non linear.

This is the case of the landing touch down spectrum. For this case, the landing gear damper is clearly a non-linear element and, of course, the incremental stress cannot be derived from the analysis of a unique load case, but from several ones.

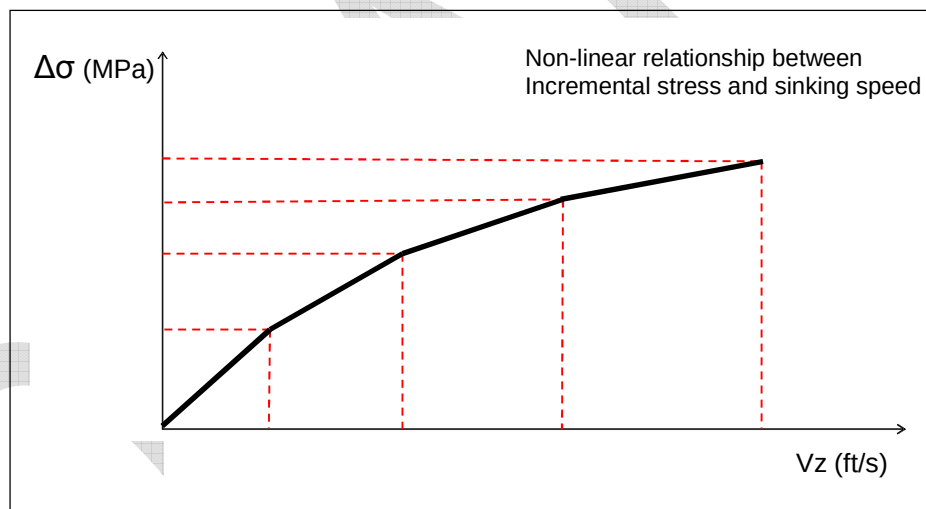


Figure 5. Example of non-linear response

In these cases, several loading conditions are needed to convert each value of the basic magnitude (for instance V_z) in the associated incremental stress $\Delta\sigma$, interpolating in the curve as shown above.

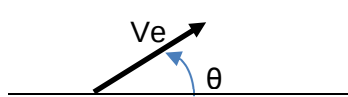
2.2.2 Combined gusts

The current policy (mandatory) from Airworthiness Authorities is not to consider vertical and lateral gusts separately, but to develop spectra for combined gusts.

For this reason, the first step is to find the most critical gust angle θ , that means the angle for which the incremental stress is a maximum. Once it is found, the incremental stress due to the combined gust is:

$$\Delta\sigma = \left(\frac{\partial\sigma}{\partial V_e}\right)_{vert} \cdot \sin\theta + \left(\frac{\partial\sigma}{\partial V_e}\right)_{horiz} \cdot \cos\theta$$

Where the angle θ is measured from the horizontal, anticlockwise sense.



The terms $\left(\frac{\partial\sigma}{\partial V_e}\right)_{vert}$ and $\left(\frac{\partial\sigma}{\partial V_e}\right)_{horiz}$ are the unitary stresses for vertical and lateral gusts, respectively.

The above expression, derived with respect to the angle θ and equalled to zero, provides the angle for which the incremental stress is a maximum, resulting:

$$\tan\theta = \frac{\left(\frac{\partial\sigma}{\partial V_e}\right)_{vert}}{\left(\frac{\partial\sigma}{\partial V_e}\right)_{horiz}}$$

This rule can be applied not only for discrete gusts, coming from ESDU or Airbus UK, but also for PSD (Power Spectral Density) gusts.

2.2.3 Active flight laws

Basically, there are two kind of spectra, attending to the evolution of loading:

- Controlled loads:

These kind of disturbances are promoted by actions from the pilot or from the AFCS (Automatic Flight Control System).

The loads vary smoothly from the steady condition up to the selected one, without any kind of transient loads.

Some examples are: Vertical manoeuvres, coordinated turns, ground turns, braking, etc.

- Uncontrolled loads:

In these kind of disturbances the origin of loads is random by nature and the evolution of the load state includes transient loads.

Some Examples are: Gusts, ground roll, landing touch down, etc.

In case of vertical gusts and in order to mitigate their effect, some flight laws, for instance the LAF (Load Alleviation Function), can be included in the Flight Control System. When a gust (discrete or

coming from continuous turbulence) is detected, the system acts automatically on movable surfaces to reduce the loading on the wing.

Some examples are the following:

- Airbus A330: When the systems detect a gust, it promotes a motion on the elevator.
- Airbus A400M: In this case the response of the system is a symmetric deflection of ailerons.

These features have to be considered when derived incremental stresses from the basic magnitude of the disturbance.

2.2.4 Dynamic Response Factor

Regarding again the uncontrolled load sources, like gusts, ground rolls or landing touch down, the load evolution includes transient effects.

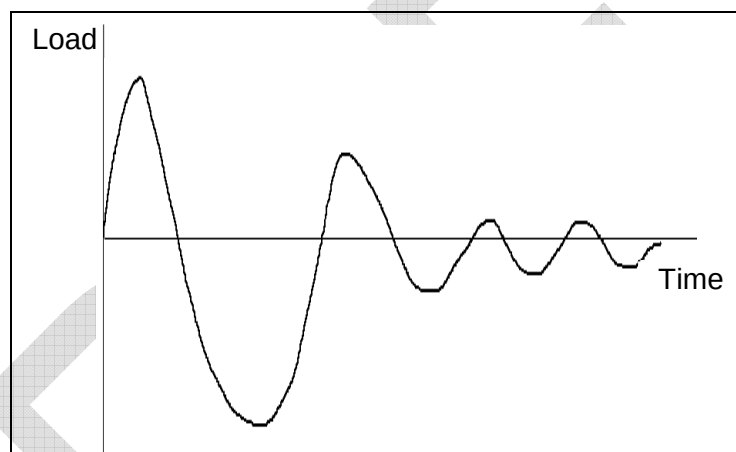


Figure 6. Time history for uncontrolled loads

Sometimes, the inclusion in the spectrum of only one cycle associated to the basic disturbance, can lead to a non-conservative estimation for later applications, like fatigue or crack growth analysis.

To cover this fact, a factor is applied on the stress spectrum frequencies. This factor, called as DRF (Dynamic Response Factor), has to be calculated for each single stress spectrum, when applicable, by considering the effect (in fatigue, for instance) of the complete time history versus the single cycle.

Although this factor can be different for each structural element, can be approached for all the locations of a subcomponent (wing, fuselage, tailplane, etc.). For this reason in Airbus D&S this factor is included into the flight profile definition and therefore directly applied on the basic spectra.

2.3 COMBINED STRESS SPECTRUM

Once de cumulative stress spectra have been obtained for all incremental loads, they can be added inside each flight segment.

The procedure to do this consists, for each stress level σ , in the addition of the cumulative frequencies associated to each single stress spectrum, $N_1, N_2, N_3 \dots$

Therefore the cumulative frequency corresponding to a stress level σ , will be:

$$N(\sigma) = \sum N_i(\sigma)$$

Next figure shows how to obtain (graphically) the combined stress spectrum from the single stress spectra:

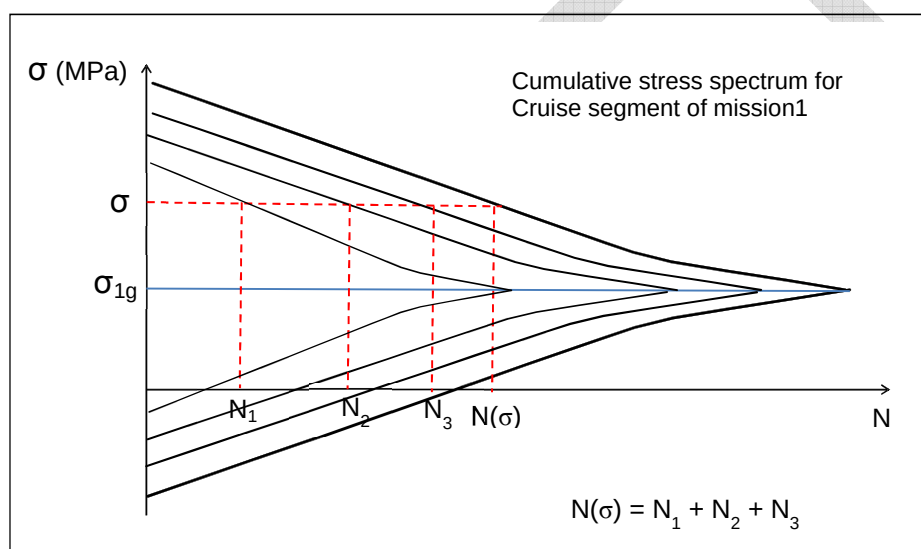


Figure 7. Example of cumulative stress spectrum

2.4 DISCRETE STRESS SPECTRUM

Finally, the global stress spectrum for each flight segment can be discretised to obtain a constants stress blocks spectrum, ready to be used for fatigue analysis purposes.

Next figure is an example showing how to discretise the above cumulative spectrum into stress blocks.

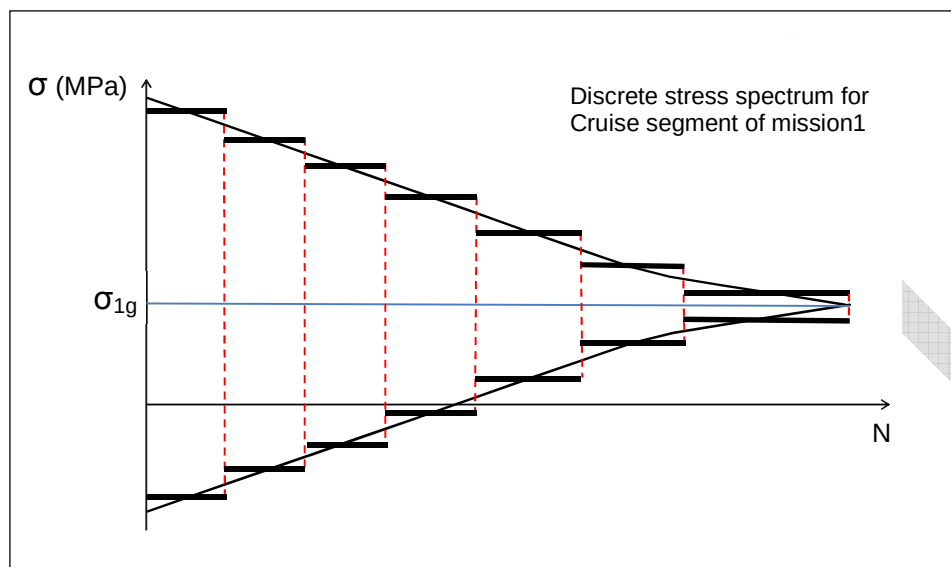


Figure 8. Example of discrete stress spectrum

Resulting for each block:

$$\sigma_{\max} = (\sigma_{\max}(i) + \sigma_{\max}(j)) / 2$$

$$\sigma_{\min} = (\sigma_{\min}(i) + \sigma_{\min}(j)) / 2$$

$$n_{\text{cyc}} = N(j) - N(i)$$

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
AFCS		Automatic Flight Control System
DRF		Dynamic Response Factor
LAF		Load Alleviation Function
σ_{1g}	MPa	Stress for Steady flight or ground condition
$\Delta\sigma$	MPa	Incremental stress due to a disturbance out from the steady condition
σ_{max}	MPa	Maximum stress of the cycle
σ_{min}	MPa	Minimum stress of the cycle
N		Number of cycles (cumulative spectrum)
n_{cyc}		Number of cycles (discrete spectrum)

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	TEAS –NT-1710004	A	Genes 3 Theoretical Manual.
Ref. 2	M00RP0411804	6.0	A400M Fatigue Load Philosophy.
Ref. 3	TAE-FG-NT-160014	A	A330-MRTT Fatigue Load Philosophy for updated F&DT analysis (INTA TC).
Ref. 4	DT 03-3205	A	CN-235-300M CG01 Spectra Development.
Ref. 5	DT-5-ADF-98003	B	C-295 Spectra Development

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 2. Software programs referenced in this chapter

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub- chapters affected
0.1 15/07/2020	M.C. Blanquer D. Fernández	First issue	All pages

Table 3. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Analysis (TEPAA)	Javier Gómez-Escalonilla DD/MM/AAAA	Holger Hickethier DD/MM/AAAA
Stress Methods (TEPAP-TL4)	Darío R. Fernández Villalba 15/07/2020	---

Table 4. *Approval signatures table for last revision of this chapter*